

- Description

This module manages an online meeting scheduling system, ensuring that:

1. **Users can book meeting slots if the requested time is available.**
2. **Overlapping meetings are not allowed**, ensuring conflict-free scheduling.
3. **Meetings have a maximum duration**, preventing excessive bookings.
4. **Users can cancel their meetings, freeing up the time slot for others.**
5. **A meeting log is maintained** for tracking all scheduled and completed meetings.

This system ensures efficient and fair allocation of meeting slots, avoiding scheduling conflicts.

- Specification

module OnlineMeetingScheduling

var meetings: sequence of (string, integer, integer)

var meetingLog: sequence of string

var maxDuration: integer

process scheduleMeeting(organizer: string, startTime: integer, duration: integer) flag: boolean

functional scenario form:

duration <= maxDuration && flag = true and meetings = meetings $\oplus$ (organizer, startTime, duration) and meetingLog = meetingLog $\oplus$ ("Scheduled: " + organizer + " from " + startTime + ":00 to " + (startTime + duration) + ":00"), Meeting scheduled successfully.

||

duration > maxDuration && flag = false, Invalid duration! Meeting request rejected.

||

conflictDetected=1 && flag = false, Meeting conflict detected! Scheduling failed.

end process

end module

- Java Code

```
package code;
import java.util.*;
class Meeting {
    private String organizer;
    private int startTime;
    private int duration;
    public Meeting(String organizer, int startTime, int duration) {
        this.organizer = organizer;
        this.startTime = startTime;
        this.duration = duration;
    }
    public String getOrganizer() {
```

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        return organizer;
    }
    public int getStartTime() {
        return startTime;
    }
    public int getEndTime() {
        return startTime + duration;
    }
}
class MeetingScheduler {
    private List<Meeting> meetings;
    private final int maxDuration;
    private List<String> meetingLog;
    public MeetingScheduler(int maxDuration) {
        this.meetings = new ArrayList<>();
        this.maxDuration = maxDuration;
        this.meetingLog = new ArrayList<>();
    }
    public boolean scheduleMeeting(String organizer, int startTime, int
duration) {
        if (duration <= 0 || duration > maxDuration) {
            System.out.println("Invalid meeting duration! Must be betwe
en 1 and " + maxDuration + " hours.");
            return false;
        }
        int endTime = startTime + duration;
        for (Meeting m : meetings) {
            if (!(endTime <= m.getStartTime() || startTime >= m.getEndT
ime())) {
                System.out.printf("Meeting conflict detected! %s's meet
ing overlaps with another meeting.%n", organizer);
                return false;
            }
        }
        Meeting newMeeting = new Meeting(organizer, startTime, duration
);
        meetings.add(newMeeting);
        meetingLog.add(String.format("Scheduled: %s from %d:00 to %d:00
", organizer, startTime, endTime));
        System.out.printf("Meeting scheduled successfully: %s from %d:0
0 to %d:00%n", organizer, startTime, endTime);
        return true;
    }
    public boolean cancelMeeting(String organizer, int startTime) {

```

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        Iterator<Meeting> iterator = meetings.iterator();
        while (iterator.hasNext()) {
            Meeting m = iterator.next();
            if (m.getOrganizer().equals(organizer) && m.getStartTime()
== startTime) {
                iterator.remove();
                meetingLog.add(String.format("Cancelled: %s's meeting a
t %d:00", organizer, startTime));
                System.out.printf("Meeting cancelled: %s at %d:00%n", o
rganizer, startTime);
                return true;
            }
        }
        System.out.println("No matching meeting found for cancellation.
");
        return false;
    }
    public void printMeetingLog() {
        System.out.println("\nMeeting Log:");
        for (String log : meetingLog) {
            System.out.println(log);
        }
    }
}

public class c15_OnlineMeetingSystem {
    public static void main(String[] args) {
        MeetingScheduler scheduler = new MeetingScheduler(4);
        // testcase-VT:
        // scheduler.scheduleMeeting("Alice", 9, 2);
        // scheduler.scheduleMeeting("Bob", 11, 2);
        // scheduler.scheduleMeeting("Charlie", 10, 2);
        // scheduler.scheduleMeeting("David", 14, 5);
        // scheduler.scheduleMeeting("Eve", 13, 2);
        // scheduler.cancelMeeting("Bob", 11);
        // scheduler.cancelMeeting("Eve", 11);
        // scheduler.scheduleMeeting("Frank", 11, 1);
        // scheduler.cancelMeeting("Hork", 11);
        // scheduler.scheduleMeeting("Hork", 11, 0);
        // scheduler.printMeetingLog();

        // testcase-FT:
        // scheduler.scheduleMeeting("maco5", 24, 1633774996);
        // scheduler.scheduleMeeting("?o5", 0, 0);
        // scheduler.scheduleMeeting("", 0, 0);
    }
}

```

```

//      scheduler.scheduleMeeting("////", 22, 791621423);
//      scheduler.scheduleMeeting("??", 0, 1766592256);
//      scheduler.cancelMeeting("////", 22);
//      scheduler.scheduleMeeting("B*", 1, 100664576);
//      scheduler.scheduleMeeting("", 10, 1697473647);
//      scheduler.scheduleMeeting("B*", 1, 100664576);
//      scheduler.scheduleMeeting("", 17, 67109206);
//      scheduler.printMeetingLog();

// testcase-Z3:
//      scheduler.scheduleMeeting("Alice", 9, -1);
//      scheduler.scheduleMeeting("Bob", 11, 11);
//      scheduler.scheduleMeeting("Charlie", 10, 10);
//      scheduler.scheduleMeeting("David", 14, 14);
//      scheduler.scheduleMeeting("David", 14, 14);
//      scheduler.scheduleMeeting("Eve", 13, 13);
//      scheduler.scheduleMeeting("Frank", 11, 1);
//      scheduler.scheduleMeeting("Frank", 11, 1);
//      scheduler.scheduleMeeting("Hork", 11, 0);
//      scheduler.printMeetingLog();

// testcase-UVT:
scheduler.scheduleMeeting("Alice", 9, 2);
scheduler.scheduleMeeting("Bob", 11, 2);
scheduler.scheduleMeeting("Charlie", 10, 2);
scheduler.scheduleMeeting("David", 14, 5);
scheduler.scheduleMeeting("Eve", 13, 2);
scheduler.scheduleMeeting("Charlie", 10, 2);
scheduler.scheduleMeeting("Eve", 13, 2);
scheduler.scheduleMeeting("Frank", 11, 1);
scheduler.cancelMeeting("Hork", 11);
scheduler.scheduleMeeting("Hork", 11, 0);
scheduler.printMeetingLog();
    }
}

```

- Coverage report

1. Test case number: 10
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	2.2 %	18	818	836
▼ src	2.2 %	18	818	836
▼ code	2.2 %	18	818	836
▼ c15_OnlineMeetingSystem.java	100.0 %	18	0	18
> c15_OnlineMeetingSystem		0	0	0
> Meeting		0	0	0
> MeetingScheduler	100.0 %	18	0	18

3. Vibration testing instruction coverage: 99%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.8 %	291	15,929	16,220
▼ src	1.8 %	291	15,929	16,220
▼ code	1.8 %	291	15,929	16,220
▼ c15_OnlineMeetingSystem.java	99.0 %	291	3	294
> c15_OnlineMeetingSystem	95.6 %	65	3	68
> Meeting	100.0 %	24	0	24
> MeetingScheduler	100.0 %	202	0	202

4. Fuzz testing branch coverage: 27.8%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	831	836
▼ src	0.6 %	5	831	836
▼ code	0.6 %	5	831	836
▼ c15_OnlineMeetingSystem.java	27.8 %	5	13	18
> MeetingScheduler	27.8 %	5	13	18
> c15_OnlineMeetingSystem		0	0	0

5. Fuzz testing instruction coverage: 40.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	121	16,101	16,222
▼ src	0.7 %	121	16,101	16,222
▼ code	0.7 %	121	16,101	16,222
▼ c15_OnlineMeetingSystem.java	40.9 %	121	175	296
> MeetingScheduler	26.7 %	54	148	202
> c15_OnlineMeetingSystem	95.7 %	67	3	70

6. Z3 branch coverage: 55.6%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	830	840
▼ src	1.2 %	10	830	840
▼ code	1.2 %	10	830	840
▼ c15_OnlineMeetingSystem.java	55.6 %	10	8	18
> MeetingScheduler	55.6 %	10	8	18
> c15_OnlineMeetingSystem		0	0	0
> Meeting		0	0	0

7. Z3 instruction coverage: 76.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.0 %	223	21,867	22,090
▼ src	1.0 %	223	21,867	22,090
▼ code	1.0 %	223	21,867	22,090
▼ c15_OnlineMeetingSystem.java	76.6 %	223	68	291
> MeetingScheduler	69.3 %	140	62	202
> Meeting	87.5 %	21	3	24
> c15_OnlineMeetingSystem	95.4 %	62	3	65

8. Unimproved V-method branch coverage: 77.8%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.7 %	14	826	840
▼ src	1.7 %	14	826	840
▼ code	1.7 %	14	826	840
▼ c15_OnlineMeetingSystem.java	77.8 %	14	4	18
> MeetingScheduler	77.8 %	14	4	18
> c15_OnlineMeetingSystem		0	0	0
> Meeting		0	0	0

9. Unimproved V-method instruction coverage: 85.5%

Element		Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼  VibrationTestingTool		1.1 %	253	21,842	22,095
▼  src		1.1 %	253	21,842	22,095
▼  code		1.1 %	253	21,842	22,095
▼  c15_OnlineMeetingSystem.java		85.5 %	253	43	296
>  MeetingScheduler		80.2 %	162	40	202
>  c15_OnlineMeetingSystem		95.7 %	67	3	70
>  Meeting		100.0 %	24	0	24